

No One Knows the State of the Art in Geospatial Foundation Models

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github.com/taylor-geospatial/gfm-leaderboard

Abstract

Geospatial foundation models (GFMs) have been proposed as generalizable backbones for disaster response, land-cover mapping, food-security monitoring, and other high-stakes Earth-observation tasks. Yet the published work about these models does not give reviewers or users enough information to tell which model fits a given task. **We argue that nobody knows what the current state of the art is in geospatial foundation models.** The methods may be useful, but the GFM literature does not standardize evaluations, training and testing protocols, released weights, or pretraining controls well enough for anyone to compare or rank them. In a 152-paper audit, we find 46 cross-paper disagreements of at least 10 points for the same model, benchmark, and protocol; 94/126 papers with extractable pretraining data use a configuration no other paper uses; and 39% of GFM papers release no model weights. This lack of community standards can be solved. We propose six concrete expectations: named-license weight release, shared core evaluations, copied-versus-rerun baseline annotations, variance reporting, one shared evaluation harness, and data-vs-architecture-vs-algorithm controls. These gaps are a coordination failure, not a fault of any individual lab; the authors of this paper, like many others in the GFM community, have contributed to them. Rather than just critiquing the community, we aim to provide concrete steps toward a shared understanding of how to innovate GFMs.

1 Introduction

The promise of geospatial foundation models is cheap, easy reuse across domains. A single pretrained Earth-observation backbone should transfer across sensors, geographies, label regimes, and downstream tasks: crop mapping, flood mapping, building extraction, forest monitoring, land-cover change, and more. That promise makes evaluation harder than ordinary model comparison. A paper may compare M models across N benchmarks, but the benchmarks differ in spatial resolution, modality, class definitions, geographic coverage, label quality, and whether the reported metric measures accuracy on small image patches or the quality of an actual map a user would rely on. This mirrors the benchmark-lottery problem described in broader ML evaluation work [17]: benchmark choice can dominate apparent progress when communities lack shared protocols. The GFM community therefore needs *clearer standards on how to test and compare GFMs*.

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Bommasani et al. [5] introduced the term for models trained on broad data that can be adapted to many downstream tasks; BERT [19], GPT-3 [10], CLIP [56], DINO [11], SAM [39], ImageNet-pretrained [18] ResNets [30], and ViTs [21] became useful partly because other groups could evaluate, reuse, and build on them. The trend also quickly swept the geospatial community: we identify 152 papers (2019–2025) in our audited corpus self-identifying as “foundation models”. We do not relitigate who may use that title. We ask a narrower question a reviewer or downstream user should be able to answer from the published record: which GFM is most performant across diverse or particular tasks by comparable empirical evidence?

Right now, the answer is unclear. For example, we find two papers that report Scale-MAE’s linear-probed accuracy on NWPU-RESISC45 as 33.0 and 89.6 using the same released model checkpoint and nominal protocol (§3.3). At most one can be right, and possibly neither; a reader deciding whether to use Scale-MAE cannot tell from these papers which number to trust. We document 46 such ≥ 10 -point disagreements between papers on the same model and benchmark in §3.3. This paper addresses this comparability problem and lays out how the community can fix it.

Position
Nobody knows the current state of the art for geospatial foundation models. Across the audited corpus, papers do not share a robust evaluation framework. Papers report different numbers for the same model on the same benchmark under the same nominal protocol. Architectural changes are bundled with pretraining data changes, with no ablations that fix one and vary the other. The community can and should come together to fix this. Better shared standards will help drive comparability and ultimately innovation that results in better outcomes for end-users in diverse application areas.

Scope. The foundation-model title is imported from NLP and computer vision, so it should carry the same minimum standard of evidence that made the title useful there [5]. We are not claiming that pretrained satellite-imagery backbones are useless; the gap is in how the scientific literature reports and compares them, not in whether the methods work. We do not require every GFM to use public or identical pretraining data; private and diverse data sources are compatible with foundation models when the paper treats data choice as an explicit variable. We also take no position on who deserves to call a model a foundation model. Our scope is the academic, open-source GFM literature, where public comparability is the main concern. Throughout, “the field” refers to GFM literature and its research community, not every remote-sensing or operational geospatial-ML effort.

Contributions. (1) We release a 152-paper systematic review with structured per-paper metadata (§2). (2) We describe three troubling trends in GFM papers, following Lipton and Steinhardt [46]’s argument that ML papers should make clear what caused an improvement rather than leave readers to guess. (3) We give six recommendations for authors, reviewers, venues, and benchmark maintainers in §4, designed to address the troubling trends this paper identifies. We label them **R1** through **R6**.

2 Publication corpus

To ground our position on GFM comparability in a transparent, reproducible corpus, we construct an audited collection of relevant papers. The supplementary repository contains the paper list, extraction schema, normalized tables, and scripts used for all reported number.

review and manual human review is performed for validation. The extraction prompt and validation steps are documented in Appendix B; the code and intermediate outputs are included in the supplementary materials. 46% of the 152 papers explicitly call their proposed model a foundation model in the title, abstract, or contributions. The remaining papers are earlier self-supervised remote-sensing models that prior GFM surveys include alongside more recent foundation-model papers. We include both types of papers; the 46% figure is purely descriptive.

We exclude 2026 papers (the year was incomplete at submission) and paywalled or metadata-poor venues where structured metadata could not be harvested at scale. We also exclude papers from a broader search that surfaced several hundred additional candidates, most of which released no weights, code, or pretraining data. This suggests they were not intended for reuse. Including such papers would likely move the headline numbers even further in the same direction.

Appendix B documents the full extraction and validation pipeline. All analyses of the corpus in this paper are reproducible from the released code.

3 Troubling Trends in GFM Comparisons

Three analyses follow, each with a clear claim, a comparison to a more mature subfield, and a paired recommendation (§4). We call these *troubling trends* because they are not one-off mistakes; they are repeated reporting choices that make model claims harder to understand, echoing the concerns raised by Lipton and Steinhardt [46]. Section 4 turns these trends into actions for authors, reviewers, and the community, previewed in boxes throughout this section.

3.1 Model weights are not published

Across our publication corpus, 39% of papers release no model weights. This is the minimum precondition for downstream reuse and comparison. Another 19% ship a public code repository with no released model artifact, so reuse would require attempting to retrain the model with the authors' codebase (App. C). Lack of published model weights is the first troubling trend: before the GFM community can compare models on shared benchmarks or rerun baselines, the model files have to be public.

R1. Release weights under a named license (§2, §3.5)

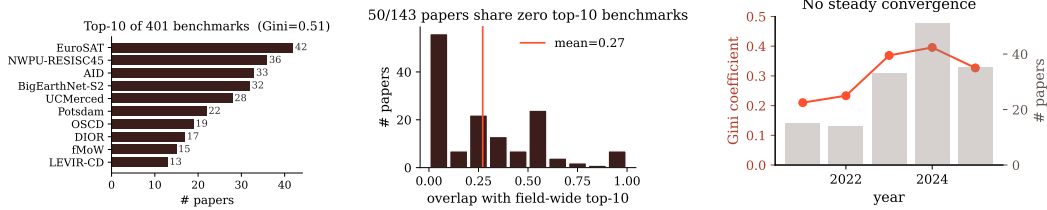
A pretrained-satellite-imagery model that is meant for reuse should release weights under a named license by camera-ready publication, or explicitly name the constraint that prevents release.

3.2 The field does not have a shared set of core benchmarks

The corpus does not converge on a shared set of benchmarks. The 152 papers in our corpus report evaluations on 401 distinct benchmarks. We determined the number of distinct benchmarks by merging benchmark aliases and excluding evaluations on auxiliary label sources such as the USDA Cropland Data Layer (see full criteria in Appendix B). The corpus has a total of 1,046 evaluation experiments, with an average of 2.6 evaluations per benchmark. The three most-used benchmarks (EuroSAT [31], NWPU-RESISC45 [13], AID [72]) together account for only 10.6% of all evaluations (Figure 1a); the remaining 89.4% is spread over 398 benchmarks, most appearing in only one or two papers. The Gini coefficient—a $[0, 1]$ measure of inequality where 0 indicates evenly distributed usage across benchmarks and 1 indicates that a single benchmark dominates—is 0.51 (95% bootstrap CI $[0.45, 0.57]$).

Heavy use of a few benchmarks would be normal: research communities converge on canonical ones. The problem is on the per-paper side. For each paper, we compute the fraction of its downstream benchmarks that overlap the top-10. The mean is 0.27 (95% CI $[0.23, 0.32]$), and 50/143 papers (35%) have zero overlap with the top-10 (Figure 1b). This shows the community is not converging on shared benchmarks over time: the year-by-year Gini coefficient is stable after 2023 (Figure 1c), and the count of benchmarks that appear in only one paper grew from 13 in 2022 to 98 in 2025.

This means no GFM in the corpus can credibly claim a literature-wide ranking from the published record: the numbers needed to rank them are not reported on enough shared benchmarks, under fixed protocols, for a fair comparison. The GFM literature has its own benchmark-lottery problem, and



(a) **Top-12 benchmarks of 401 ranked by paper count.** The top-3 (EuroSAT, NWPU-RESISC45, AID) cover only 10.6% of all evaluations; the remaining 89.4% scatters across 398 benchmarks that mostly appear in one or two papers. (b) **Histogram of per-paper overlap with the GFM top-10 benchmarks.** Mean overlap is 0.27 and 50/143 papers (35%) sit in the left-most bin at zero overlap, so more than a third of the corpus shares no evaluation ground with the most-used benchmarks. (c) **Gini coefficient of benchmark usage by publication year** (0 means evenly spread, 1 means a single benchmark dominates). Gini does not steadily increase after 2023, so the field is not converging on a shared core over time.

Figure 1: **How the 152-paper corpus uses benchmarks.** Panel (a) shows the top-10 benchmarks evaluated in the corpus; panel (b) shows that 35% of papers do not test on the most-used benchmarks at all; panel (c) shows that this pattern is not improving over time. Together the three panels say no GFM in the corpus can be ranked literature-wide, because the numbers needed for a fair comparison are not reported on enough shared benchmarks.

without the kind of community coordination that has begun to make computer-vision benchmarks more comparable [40, 54], it is unlikely to fix itself.

R2. Report on a shared core set of evaluation datasets (§3.2)

Authors making any pretrained-satellite-imagery comparison should report on a shared set of core evaluations with a clear protocol, plus extra tests for other aspects of novelty claimed.

3.3 Reported metric values diverge by tens of points across papers, at fixed protocol

A field that shares benchmarks should at least agree on reported metric values for the same model-benchmark-protocol tuple. We mined every (model, benchmark, metric, evaluation-protocol, train-regime) tuple in the 152-paper corpus (10,817 results after benchmark and metric normalization) and bucket by protocol (finetune, linear probe, kNN probe, zero-shot, few-shot). We then drop generic and classical-ML baselines (random init, ImageNet-supervised, MLP, from-scratch, LightGBM, XGBoost, SVM, kNN). We also drop detection benchmarks (DOTA [73], DIOR [43], FAIR1M [66]) whose shared mAP metric name conflates DOTA-style oriented-box mAP with COCO-style AP[.5:.95] on horizontal boxes. Every remaining disagreement is between papers within a fixed protocol bucket. After these filters, 301 tuples are reported by ≥ 2 papers, and we measure the spread (max-min) of the reported metric on each.

Of the 301 multi-paper tuples, 76 have spread ≥ 5 pts, 46 have spread ≥ 10 pts, and 20 have spread ≥ 20 pts (Figure 2, left). **The largest spread is 56.6 pts:** Scale-MAE on NWPU-RESISC45 under linear probe is reported as accuracy 33.0 by [44] and 89.6 by the original authors [57], on the same released ViT-L checkpoint under the same nominal linear-probe protocol; neither paper describes the recipe for fitting a linear-probe (i.e., details on the optimizer, head LR, or eval crop). Another example is GPT-4o on UCMerced, where zero-shot spans 43.5 [32] $\rightarrow$ 88.8 [63] ($\Delta = 45.3$ pts, $n=2$), and neither paper discloses in detail the prompting hyperparameters used. The top-10 disagreements are plotted in Figure 2 (right). These disagreements are not isolated outliers. Most multi-paper tuples agree closely (median spread ~ 0 pts), but the **90th-percentile spread is 12.7 pts**, an order of magnitude larger than typical seed variance for classification heads under fixed protocols [8, 7]. Variance for segmentation and regression decoders is rarely reported and remains an open gap.

Several plausible failure modes are consistent with the spread. Papers may copy numbers from other papers’ tables without annotating that the source used a different train/val/test split, sensor channel set, class set, normalization, or adaptation recipe. Papers may rerun baselines with less generous sweeps than the original source and report the rerun as if it were the same protocol. Vision-language rows

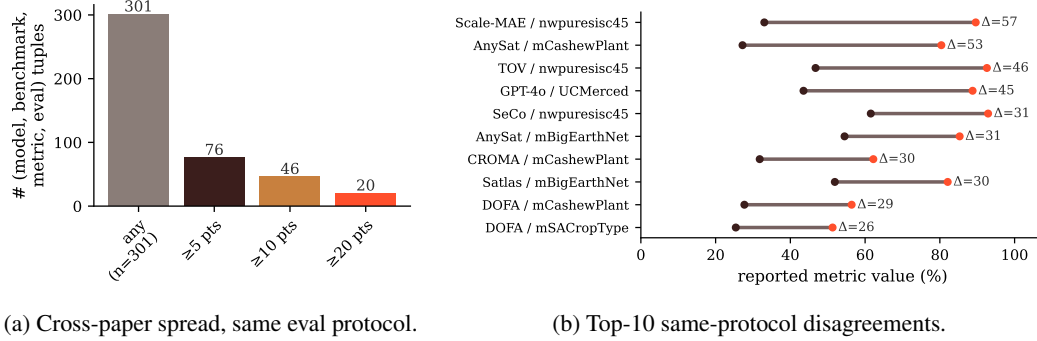


Figure 2: **Papers report wildly different numbers for the “same” experiment.** Across 301 cases with matching (model, benchmark, metric, protocol), many disagreed by ≥ 5 , ≥ 10 , or ≥ 20 points (left); the 10 largest gaps are shown right. The worst: Scale-MAE on NWPU-RESISC45 linear probing, 33.0 vs. 89.6 from the *same checkpoint and nominal setup*. Training stochasticity is ~ 1 point, so these differences are far larger than what would be expected from run-to-run variation, indicating **direct cross-paper comparisons are unreliable**.

add prompt templates, verbalizers, API snapshots, and temperature as hidden axes. In every case, the label “EuroSAT accuracy under linear probe” provides less determinism than readers assume. Table 1 (Appendix D) lists the top-10 most divergent tuples with references to the reporting papers named, so readers can refer to the source tables directly. Fuller [25]’s “BAD TABLES” talk catalogs the same confound at the architecture level: patch size, image size, channel groupings, and pretraining schedule each shift downstream accuracy by tens of points across rows that share a model name.

R3. Annotate every copied baseline number (§3.3)

Every result-table entry should be marked as either $\odot$ RERUN, with the configuration disclosed, or $\square$ COPIED, with the source paper and source protocol cited.

R4. Report seed variability on headline comparisons (§3.3)

Headline comparisons should show how stable the result is when that is practical: repeated-run mean $\pm$ std for affordable evaluations, or a clear note that the result comes from one run when repeats are too expensive. This is especially important for benchmarks where results are known to vary greatly across random seeds.

3.4 Aggregated benchmarks provide dataset bundles, not evaluation harnesses

The LLM community converged on *evaluation harnesses*, not just benchmark collections. For example, `lm-evaluation-harness` [27] is the canonical tool that powers the Open LLM Leaderboard [23]: a single Python package every model owner runs, with versioned task definitions (e.g., MMLU v0.0 vs. v0.1), reference protocol implementations, automated submission checks through continuous integration, and a common task-config format that lets any third party reproduce a reported number from a model name and a task tag. HELM [45] provides multi-metric evaluation across 87 scenarios under a continuously hosted leaderboard. BIG-bench [64] ships 200+ tasks under a unified API. Computer vision also has common harnesses: outside of ImageNet [18], VTAB [77] defines a 19-task transfer-learning protocol with reference implementations, but there is no continuously hosted, CI-gated leaderboard at the same level.

The geospatial domain lacks this infrastructure. What the GFM community has are *dataset bundles*: GitHub repositories with curated task splits, reference dataloaders, and example training scripts. GEO-Bench [41, 62] provides fixed splits and a public toolkit. PANGAEA [51] provides a unified codebase that runs encoders across a fixed task list. FoMo-Bench [6] curates a forest-monitoring task list. PhilEO Bench [22] is a paper proposing a task list, with no released harness code at the time of writing. TorchGeo [65] has broad dataset and transform coverage for geospatial ML, but it is a general-purpose library rather than a CI-gated probing harness with canonical GFM submissions.

None of these are evaluation harnesses like those available for LLMs. Each is a self-contained repository or toolkit where a researcher writes their own training loop on curated splits; cross-paper protocols are not versioned, submissions are not CI-gated, and there is no canonical tool the whole community runs.

We argue that disagreements in §3.3 occur even on shared benchmarks because there is no common evaluation harness. Even when two GFM papers evaluate on the same dataset from the same bundle, the results remain incomparable as long as evaluation protocol details such as optimizer choice, head learning rate, eval crop, and Jaccard-averaging scheme (macro vs. weighted-per-class IoU) are not consistent. Curating more datasets does not close that protocol gap. What the GFM literature is missing is a third-party evaluation harness: a versioned, openly maintained tool that every model owner runs to produce a reported number, and that any reviewer can rerun end-to-end from a model identifier.

A harness is necessary but not sufficient. Existing remote-sensing datasets often reflect where labels were convenient to collect, not the full distribution of operational tasks, geographies, sensors, and label policies. Patch-level or image-level scores also may not predict map-level accuracy or user-facing utility. Spatial autocorrelation between training and validation samples can inflate remote-sensing accuracy estimates [36, 37], so an evaluation that is reproducible can still be the wrong evaluation for a deployment claim. The right target is therefore not any one specific benchmark (EuroSAT is just a common example); it is a shared core for sanity-checking model comparisons, plus EO-native extension axes for sensor, modality, temporal, geographic, label-quality, and map-level claims.

R5. Create a shared third-party evaluation harness (§3.4)

Community benchmark maintainers and venues should build one shared evaluation tool that every model owner can run, with fixed task definitions and automatic submission checks.

3.5 Architecture and pretraining-data improvements are confounded

When a paper changes both the model and the pretraining data, readers cannot tell which change caused the gain unless one is held fixed. This is an attribution problem, not an argument for identical pretraining data. A foundation model does not need to share pretraining data – private, proprietary, and diverse pretraining data are acceptable when a novel pretraining dataset is the claimed contribution or when a paper does not ask readers to attribute gains to architecture alone. The problem in the GFM corpus is that methodological claims (architectural changes, new self-supervised objectives) are often impossible to separate from pretraining-data changes without an ablation that fixes one and varies the other. Corley et al. [15] show that apparent GFM gains over supervised ImageNet baselines on BigEarthNet shrink or vanish once the pretraining and downstream distributions are held constant: an existence proof that the architecture-vs-pretraining-data confound can hide effects the corpus would benefit from disentangling. Kaur et al. [38] shows that different pretraining datasets can shift downstream accuracy by margins comparable or larger than gains attributed to architectural novelty.

The aggregate numbers across our 152 papers make this pretraining-data gap concrete. After merging pretraining dataset aliases and dropping unnamed or misnamed pretraining datasets (see full filter list in Appendix B), we count 87 distinct named primary pretraining datasets across 126 papers that name a specific dataset (the remaining 26 papers describe their pretraining data only generically). Some papers pretrain on a single canonical dataset, but many build a mixture from multiple sources and give it a single name. For example, RS5M [80] is built from LAION [60] + CC3M [61] + CC12M [12] + others. AnySat’s GeoPlex [2] wraps TreeSatAI-TS [1] + FLAIR [28] + PLANTED [55] + PASTIS-HD [1]. GeoPile [52] wraps MillionAID [48] + SEN12MS [59] + MDAS [33]. In these

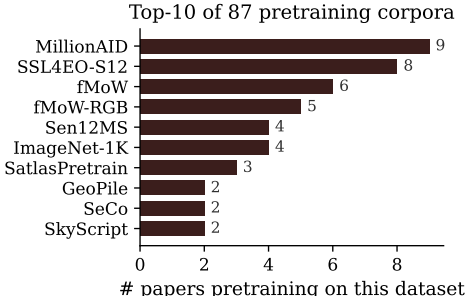


Figure 3: **Top-10 (of 87) named pretraining datasets** across the 126 corpus papers that name one. MillionAID leads at just 9 papers (~5.9% of 152); SSL4EO-S12 (8), fMoW (6), and fMoW-RGB (5) follow.

cases, we increment the count of the individual datasets, not the wrapper. Even so, the most-used primary pretraining dataset, MillionAID [48], appears in only 9 papers, followed by SSL4EO-S12 [71] (8), fMoW [14] (6), and fMoW-RGB (5) (Figure 3). The actual scene composition behind named sensor labels (Sentinel-1/2, Landsat, NAIP) is split across dozens of overlapping derived pretraining datasets (SSL4EO-S12 [71], fMoW-Sentinel [14], MMEarth [53], MajorTOM-Core [24], SatlasPretrain [3]) whose intersection cannot be audited from the papers alone.

The counts of named pretraining datasets understate the comparability gap. Two papers that name the same pretraining dataset may still pretrain on different data: a paper that pretrains on BigEarthNet alone is not equivalent to one that pretrains on a custom mixture in which BigEarthNet is one source among many. Both list BigEarthNet, but the resulting models see different data. We compute the *full* pretraining set for each paper (the deduplicated set of all named datasets the paper pretrains on), and compute how often two papers’ full sets are identical. Of 126 papers with an extractable pretraining dataset, only 32 (25%) share their full configuration with at least one other paper. The other 94 papers each pretrain on a configuration no other work uses. The largest comparability cluster is 7 papers that pretrain on MillionAID alone; the next is 6 papers on SSL4EO-S12 alone. Outside those handful of clusters, no two papers in the corpus have run the same pretraining recipe.

R6. Disentangle pretraining dataset from architecture and algorithm (§3.5)

Any new GFM that changes more than one of *a) pretraining data, b) architecture (including subtle choices like patch size), and c) algorithm* should include controlled comparisons that isolate the claimed change. Without these controls, weak results can hide behind stronger changes, making it unclear which modification actually produced the improvement.

4 Recommendations

The recommendation boxes in §3 turn each trend into an action: release reusable weights, evaluate on a shared core, distinguish copied from rerun baselines, report uncertainty, separate model changes from data changes, and build a shared evaluation tool. Most of these recommendations are actions that authors can (and, we argue, should) start doing today. We describe each recommendation below.

R1: release reusable weights or name the constraint. A model intended for reuse should ship weights under a named, permissive-by-default license by camera-ready publication. 39% of the corpus currently releases no weights, which blocks reuse and the cross-paper checks R3 and R6 need. When sensor licensing, data residency, export control, or partner restrictions prevent release, authors should name the reason rather than leave readers guessing.

R2: evaluate on a minimum set of shared core benchmarks. Authors should evaluate on a shared core set of benchmark tasks with clearly stated protocols, then add tests for the task families their model claims to handle: classification, segmentation, change detection, regression, time series, multimodal/SAR, and map-level evaluation when the claim is map-level. Initially, authors could prioritize the most common benchmarks in our corpus (see Figure 1a). Alternatively (or in addition), authors could prioritize the GEO-Bench and PANGAEA benchmark collections. We acknowledge that older or more popular benchmarks are not automatically the right ones. The GFM research and end-user community should convene to identify the core benchmarks to prioritize in evaluations.

R3: mark copied versus rerun baselines. Every baseline entry should be marked as ☐ COPIED or ☉ RERUN. Copied rows need the source paper and source protocol; rerun rows need the new configuration. Tables in which the proposed method is rerun but all baselines are copied without protocol notes are not fair comparisons. If a rerun is far from the source number, the paper should flag that rather than quietly replace the old result.

R4: report uncertainty where it is affordable. Whenever possible (e.g., for cheap heads and headline classification comparisons), authors should report mean±std over repeated runs [8]. For expensive segmentation, regression, time-series, or map-level runs, a narrower first step is enough: say whether the headline result is from one run, name the main sources of randomness, and avoid claiming a clear improvement unless the gain is likely bigger than normal run-to-run differences. Computer-vision and ML reproducibility work has shown that small implementation and seed choices can change reported gains [7], and the GFM literature has not yet measured its own run-to-run variation well enough to ignore this concern.

R5: create a shared third-party evaluation harness. The missing community layer is a third-party-maintained harness with versioned protocols, a common submission format, and automated checks that any model owner runs to produce a reported number (§3.4). GEO-Bench [41, 62], PANGAEA [51], and related dataset bundles are the starting point; TerraTorch [29] and PANGAEA take steps in this direction, but the GFM literature still lacks one shared, CI-gated tool that runs the same protocol for everyone.

R6: separate model changes from pretraining-data changes. If a paper introduces both a new pretraining dataset and a new architecture or self-supervised objective, it should include at least one comparison that fixes the data source to a public choice (e.g., SSL4EO-S12, MillionAID, fMoW, or MajorTOM-Core) at a similar token or image budget. At minimum, papers using the same sensor should compare within that sensor family. Variation in pretraining data is welcome; it should be the thing being studied, not a hidden reason for a gain [38]. The ablation does not need to be a frontier-scale run. Re-pretraining a ViT-L on a billion-image dataset is beyond many labs, and R6 should not be read as “every paper must pretrain on the same handful of datasets.” The practical standard is narrower: if the paper claims an architectural or objective improvement, include at least one controlled run on a canonical public pretraining dataset or a documented, deduplicated, region-balanced subset.

For conferences, workshops, program committees, and the community at large. Machine learning venues (e.g., ICML, ICLR, NeurIPS, CVPR, ICCV, ECCV) and geospatial-specific venues (e.g., EarthVision at CVPR, IGARSS, and ISPRS) should treat R1–R6 as default expectations and non-compliance as a substantive review concern. Reviewers cannot verify artifacts that are only promised for camera-ready, so venues should require anonymous weights, code, or executable evaluation artifacts at submission when those artifacts support the paper’s central claims; otherwise, the paper should state that the numbers cannot be independently checked during review. A reviewer need not require all six recommendations to be met. The key question is which missing check confounds the paper’s claim. We encourage reviewers to use the following checklist to help identify issues rather than treating them as a pass/fail checklist.

Reviewer’s checklist (mapped to R1–R6)

- ☒ **C1 [R1], released model files:** Are weights shared under a named license by camera-ready, or does the paper clearly state why they cannot be released?
- ☒ **C2 [R2], shared evaluation:** Are there at least three benchmarks from a shared core set (for example, the 10 most-evaluated benchmarks across the corpus, or a common bundle like GEO-Bench or PANGAEA) and is the protocol stated explicitly?
- ☒ **C3 [R3], result source:** Is every baseline row marked $\odot$ RERUN with the configuration disclosed, or $\square$ COPIED with the source paper and its protocol cited? For headline benchmarks, does the paper match a published prior number under an explicit protocol or rerun end-to-end?
- ☒ **C4 [R4], uncertainty reporting:** Does the headline table report mean $\pm$ std over ≥ 3 seeds where feasible, or clearly state that the result is single-run?
- ☒ **C5 [R6], pretraining-data control:** If the paper introduces both a new architecture or self-supervised objective *and* a new pretraining dataset, is there a comparison fixing the pretraining dataset to a shared public choice?

5 Alternative views

“The field is young; concentration and divergence will self-correct.” The current GFM survey literature [49, 74] takes this view implicitly, and the GEO-Bench stewards [41, 62] take the stronger version that a consolidating leaderboard is the self-correction vehicle. We agree that the leaderboard route is the right one. However, the failures we measure (56.6-pt within-model divergence, 35% zero top-10 overlap, 75% of papers pretraining on a unique configuration) will not fix themselves with time alone. A new model can meet R1–R6 at release, and Figure 1c shows benchmark concentration has not improved since 2023. The risk of waiting is that GFM papers keep reporting progress without knowing what drove it: the model design, the data, the protocol, run-to-run noise, or the benchmark choice. A shared standards document gives authors and reviewers a common reference before informal habits become hard to change.

“Geospatial data and tasks are heterogeneous; do not shoehorn remote sensing into a CV or LLM box.” Rolf et al. [58], Marsocci et al. [51], Fuller [25] argue that input heterogeneity (multispectral, SAR, multi-resolution, multi-temporal) should set remote-sensing-native standards. We are not arguing for one mandated input format. Instead we argue that input-axis heterogeneity itself does not explain disagreement within an axis. The Scale-MAE/RESISC45 disagreement (§3.3) is on a single benchmark, single protocol, single checkpoint. We ask for controls plus diversity: a small shared core, with explicit extension axes for sensor-, modality-, and time-series-specific tasks.

“These problems are driven by bad actors, not the whole field.” Our corpus already removes some noise by excluding metadata-poor venues and obvious non-reuse papers, so the trends are not driven by an anything-goes scrape. Still, we acknowledge that a stricter “top papers only” check could show different trends, but there is no obvious filter for this (e.g., citations? year? author names?). Our claim is about the public record a reviewer, leaderboard maintainer, or downstream user actually sees. Those readers should not need informal field knowledge to know which papers are careful, which baselines were copied, or which protocol was rerun. If the reliable signal exists only inside a small social circle, it is not yet serving as public evidence.

“Model weights or pretraining data sometimes can’t be open-sourced because of private or proprietary data constraints.” We agree that proprietary models that are closed-weight or closed-data can still add value to the research community; AlphaEarth [9] is one such example in geospatial foundation models. R1 recommends that authors state such constraints if they are blocking the release of model weights or pretraining data, but if such constraints are not present, all artifacts should be released. We do, however, contend that open-source practices are largely beneficial for the goals stated in this paper, in line with Donoho’s notion of “frictionless reproducibility” [20].

6 Conclusion

We argue that nobody knows the current state of the art for geospatial foundation models because the published literature does not yet share the controls a reader needs to compare them. The evidence from our 152-paper audit supports this claim: papers do not share enough benchmarks, the same model on the same benchmark gets very different reported scores across papers, and pretraining setups vary so much that model gains are hard to attribute without controls. We suggest six recommendations for authors and the research community to correct the troubling trends identified in this paper, focusing on actions authors can take now or in the near term. We also suggest a reviewer checklist to help identify when the experiments or contributions in a GFM paper are cause for concern.

Limitations. Our analysis relies on LLM-based automated extraction, public APIs, and public data. Initial extraction of structured metadata was performed using Docling for PDF sources and Claude Opus 4.7 & GPT 5.5 Codex for LaTeX sources, followed by manual human verification of all extracted fields against the source papers. Human error is still possible. Details of the extraction are described in App. B.

What the future looks like if we are successful. A user who needs a GFM for crop mapping, flood response, or building detection can open a maintained leaderboard, choose the task, sensor, region, and date range they care about, and trust that the top entries were run under the same protocol rather than stitched together from incomparable papers. A researcher releasing a new model does not need to compare against every backbone from the last seven years just because nobody knows which few baselines still matter; they run the shared harness, mark any extra copied numbers as ☐ COPIED, and spend the rest of the paper explaining what actually changed. A reviewer can ask whether a missing check changes the claim, not whether the whole table is real. A leaderboard maintainer can rerun a result from a model identifier, see when a protocol changed, and flag scores whose uncertainty or data controls are missing. In that future, GFMs can still be diverse in architecture, scale, modality, and application, but their public evidence becomes boring in the best way: comparable enough that the interesting arguments are about models and data, not table archaeology. *To know which geospatial foundation model is best, we must first make them comparable.*

Acknowledgments

We thank Konstantin Klemmer for his careful and in-depth review of the manuscript. His detailed comments substantially improved the clarity and presentation of this work.

References

- [1] Guillaume Astruc, Nicolas Gonthier, Clement Mallet, and Loic Landrieu. Omnisat: Self-supervised modality fusion for earth observation. In *European Conference on Computer Vision*, pages 409–427. Springer, 2024.
- [2] Guillaume Astruc, Nicolas Gonthier, Clement Mallet, and Loic Landrieu. Anysat: One earth observation model for many resolutions, scales, and modalities. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 19530–19540, 2025.
- [3] Favyen Bastani, Piper Wolters, Ritwik Gupta, Joe Ferdinando, and Aniruddha Kembhavi. Satlaspretrain: A large-scale dataset for remote sensing image understanding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 16772–16782, 2023.
- [4] Favyen Bastani et al. Olmoeearth: Stable latent image modeling for multimodal earth observation. *arXiv preprint*, 2025.
- [5] Rishi Bommasani, Drew A. Hudson, Ehsan Adeli, et al. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*, 2021.
- [6] Nikolaos Ioannis Bountos, Arthur Ouaknine, Ioannis Papoutsis, and David Rolnick. FoMo: Multi-modal, multi-scale and multi-task remote sensing foundation models for forest monitoring. In *AAAI Conference on Artificial Intelligence*, pages 27858–27868, 2025. doi: 10.1609/aaai.v39i27.35002.
- [7] Xavier Bouthillier, César Laurent, and Pascal Vincent. Unreproducible research is reproducible. In *International Conference on Machine Learning*, pages 725–734. PMLR, 2019.
- [8] Xavier Bouthillier, Pierre Delaunay, Mirko Bronzi, et al. Accounting for variance in machine learning benchmarks. In *MLSys*, 2021.
- [9] Christopher F Brown, Michal R Kazmierski, Valerie J Pasquarella, William J Rucklidge, Masha Samsikova, Chenhui Zhang, Evan Shelhamer, Estefania Lahera, Olivia Wiles, Simon Ilyushchenko, et al. Alphaearth foundations: An embedding field model for accurate and efficient global mapping from sparse label data. *arXiv preprint arXiv:2507.22291*, 2025.
- [10] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.
- [11] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 9650–9660, 2021.
- [12] Soravit Changpinyo, Piyush Sharma, Nan Ding, and Radu Soricut. Conceptual 12m: Pushing web-scale image-text pre-training to recognize long-tail visual concepts. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 3558–3568, 2021.
- [13] Gong Cheng, Junwei Han, and Xiaoqiang Lu. Remote sensing image scene classification: Benchmark and state of the art. *Proceedings of the IEEE*, 105(10):1865–1883, 2017.
- [14] Gordon Christie, Neil Fendley, James Wilson, and Ryan Mukherjee. Functional map of the world. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 6172–6180, 2018.
- [15] Isaac Corley, Caleb Robinson, and Anthony Ortiz. Revisiting pre-trained remote sensing model benchmarks: Resizing and normalization matters. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 3162–3172, 2024. doi: 10.1109/CVPRW63382.2024.00322.

- [16] Muhammad Sohail Danish, Muhammad Akhtar Munir, Syed Roshaan Ali Shah, Muhammad Haris Khan, Rao Muhammad Anwer, Jorma Laaksonen, Fahad Shahbaz Khan, and Salman Khan. Terrafm: A scalable foundation model for unified multisensor earth observation. *arXiv preprint arXiv:2506.06281*, 2025.
- [17] Mostafa Dehghani, Yi Tay, Alexey A. Gritsenko, et al. The benchmark lottery. *arXiv preprint arXiv:2107.07002*, 2021.
- [18] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pages 248–255. Ieee, 2009.
- [19] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, pages 4171–4186, 2019.
- [20] David Donoho. Data science at the singularity. *Harvard Data Science Review*, 6(1), 2024.
- [21] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.
- [22] Casper Fibaek, Luke Camilleri, Andreas Luyts, Nikolaos Dionelis, and Bertrand Le Saux. Phileo bench: Evaluating geo-spatial foundation models. In *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2024.
- [23] Clémentine Fourier, Nathan Habib, Alina Lozovskaya, Konrad Hynek, and Thomas Wolf. Open LLM leaderboard v2, 2024. https://huggingface.co/spaces/open-llm-leaderboard/open_llm_leaderboard.
- [24] Alistair Francis and Mikolaj Czerkawski. Major tom: Expandable datasets for earth observation. In *IGARSS 2024-2024 IEEE International Geoscience and Remote Sensing Symposium*, pages 2935–2940. IEEE, 2024.
- [25] Anthony Fuller. Bad tables: Why you shouldn’t trust results tables in remote-sensing foundation model papers, 2026. URL https://antofuller.github.io/BAD_TABLES.pdf. Talk, ICLR Machine Learning for Remote Sensing Workshop, April 2026.
- [26] Anthony Fuller, Koreen Millard, and James Green. Croma: Remote sensing representations with contrastive radar-optical masked autoencoders. *Advances in Neural Information Processing Systems*, 36:5506–5538, 2023.
- [27] Leo Gao, Jonathan Tow, Baber Abbasi, et al. A framework for few-shot language model evaluation. *Zenodo*, 2024. `lm-evaluation-harness`.
- [28] Anatol Garioud, Nicolas Gonthier, Loic Landrieu, Apolline De Wit, Marion Valette, Marc Poupée, Sébastien Giordano, et al. Flair: a country-scale land cover semantic segmentation dataset from multi-source optical imagery. *Advances in Neural Information Processing Systems*, 36:16456–16482, 2023.
- [29] Carlos Gomes, Benedikt Blumenstiel, Joao Lucas De Sousa Almeida, Pedro Henrique De Oliveira, Paolo Fraccaro, Francesc Marti Escofet, Daniela Szwarcman, Naomi Simumba, Romeo Kienzler, and Bianca Zadrozny. Terratorch: The geospatial foundation models toolkit. In *IGARSS 2025-2025 IEEE International Geoscience and Remote Sensing Symposium*, pages 6364–6368. IEEE, 2025.
- [30] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016.

- [31] Patrick Helber, Benjamin Bischke, Andreas Dengel, and Damian Borth. Eurosat: A novel dataset and deep learning benchmark for land use and land cover classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 12(7):2217–2226, 2019.
- [32] Huiyang Hu, Peijin Wang, Yingchao Feng, Kaiwen Wei, Wenxin Yin, Wenhui Diao, Mengyu Wang, Hanbo Bi, Kaiyue Kang, Tong Ling, et al. Ringmo-agent: A unified remote sensing foundation model for multi-platform and multi-modal reasoning. *arXiv preprint arXiv:2507.20776*, 2025.
- [33] Jingliang Hu, Rong Liu, Danfeng Hong, Andrés Camero, Jing Yao, Mathias Schneider, Franz Kurz, Karl Segl, and Xiao Xiang Zhu. Mdas: A new multimodal benchmark dataset for remote sensing. *Earth System Science Data*, 15(1):113–131, 2023.
- [34] Ziyue Huang, Mingming Zhang, Yuan Gong, Qingjie Liu, and Yunhong Wang. Generic knowledge boosted pretraining for remote sensing images. *IEEE Transactions on Geoscience and Remote Sensing*, 62:1–13, 2024.
- [35] Yuru Jia, Valerio Marsocci, Ziyang Gong, Xue Yang, Maarten Vergauwen, and Andrea Nascetti. Can generative geospatial diffusion models excel as discriminative geospatial foundation models? In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 8429–8440, 2025.
- [36] Nicolas Karasiak, Jean-François Dejoux, Claude Monteil, and David Sheeren. Spatial dependence between training and test sets: another pitfall of classification accuracy assessment in remote sensing. *Machine Learning*, 111:2715–2740, 2022. doi: 10.1007/s10994-021-05972-1.
- [37] Teja Kattenborn, Felix Schiefer, Julian Frey, Hannes Feilhauer, Miguel D. Mahecha, and Carsten F. Dormann. Spatially autocorrelated training and validation samples inflate performance assessment of convolutional neural networks. *ISPRS Open Journal of Photogrammetry and Remote Sensing*, 5:100018, 2022. doi: 10.1016/j.ophoto.2022.100018.
- [38] Amandeep Kaur, Mirali Purohit, Gedeon Muhawenayo, Esther Rolf, and Hannah Kerner. Pretrain where? investigating how pretraining data diversity impacts geospatial foundation model performance. *arXiv preprint arXiv:2604.21104*, 2026.
- [39] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C Berg, Wan-Yen Lo, et al. Segment anything. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 4015–4026, 2023.
- [40] Bernard Koch, Emily Denton, Alex Hanna, and Jacob G. Foster. Reduced, reused and recycled: The life of a dataset in machine learning research. In *NeurIPS Datasets and Benchmarks*, 2021.
- [41] Alexandre Lacoste, Nils Lehmann, Pau Rodríguez Castaño, et al. GEO-Bench: Toward foundation models for earth monitoring. In *NeurIPS Datasets and Benchmarks*, 2023.
- [42] Alexandre Lacoste, Nils Lehmann, Pau Rodriguez, Evan Sherwin, Hannah Kerner, Björn Lütjens, Jeremy Irvin, David Dao, Hamed Alemohammad, Alexandre Drouin, et al. Geo-bench: Toward foundation models for earth monitoring. *Advances in Neural Information Processing Systems*, 36:51080–51093, 2023.
- [43] Ke Li, Gang Wan, Gong Cheng, Liqiu Meng, and Junwei Han. Object detection in optical remote sensing images: A survey and a new benchmark. *ISPRS journal of photogrammetry and remote sensing*, 159:296–307, 2020.
- [44] Zhihao Li, Biao Hou, Siteng Ma, Zitong Wu, Xianpeng Guo, Bo Ren, and Licheng Jiao. Masked angle-aware autoencoder for remote sensing images. In *European Conference on Computer Vision*, pages 260–278. Springer, 2024.
- [45] Percy Liang, Rishi Bommasani, Tony Lee, et al. Holistic evaluation of language models. *Transactions on Machine Learning Research*, 2023.

- [46] Zachary C Lipton and Jacob Steinhardt. Troubling trends in machine learning scholarship: Some ml papers suffer from flaws that could mislead the public and stymie future research. *Queue*, 17(1):45–77, 2019.
- [47] Nikolaos Livathinos, Christoph Auer, Maksym Lysak, Ahmed Nassar, Michele Dolfi, Panos Vagenas, Cesar Berrospi Ramis, Matteo Omenetti, Kasper Dinkla, Yusik Kim, et al. Do-cling: An efficient open-source toolkit for ai-driven document conversion. *arXiv preprint arXiv:2501.17887*, 2025.
- [48] Yang Long, Gui-Song Xia, Shengyang Li, Wen Yang, Michael Ying Yang, Xiao Xiang Zhu, Liangpei Zhang, and Deren Li. On creating benchmark dataset for aerial image interpretation: Reviews, guidances and million-aid. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 14:4205–4230, 2021.
- [49] Siqi Lu, Junlin Guo, James R. Zimmer-Dauphinee, et al. Vision foundation models in remote sensing: A survey. *IEEE Geoscience and Remote Sensing Magazine*, 2024.
- [50] Oscar Manas, Alexandre Lacoste, Xavier Giró-i Nieto, David Vazquez, and Pau Rodriguez. Seasonal contrast: Unsupervised pre-training from uncurated remote sensing data. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 9414–9423, 2021.
- [51] Valerio Marsocci, Yuru Jia, Gilles Le Bellier, et al. PANGAEA: A global and inclusive benchmark for geospatial foundation models. *arXiv preprint arXiv:2412.04204*, 2024.
- [52] Matías Mendieta, Boran Han, Xingjian Shi, Yi Zhu, and Chen Chen. Towards geospatial foundation models via continual pretraining. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 16806–16816, 2023.
- [53] Vishal Nedungadi, Ankit Karirya, Stefan Oehmcke, Serge Belongie, Christian Igel, and Nico Lang. Mmearth: Exploring multi-modal pretext tasks for geospatial representation learning. In *European Conference on Computer Vision*, pages 164–182. Springer, 2024.
- [54] Simon Ott, Adriano Barbosa-Silva, Kathrin Blagec, Jan Brauner, and Matthias Samwald. Mapping global dynamics of benchmark creation and saturation in artificial intelligence. *Nature Communications*, 2022.
- [55] Luis Miguel Pazos-Outón, Cristina Nader Vasconcelos, Anton Raichuk, Anurag Arnab, Dan Morris, and Maxim Neumann. Planted: a dataset for planted forest identification from multi-satellite time series. In *IGARSS 2024-2024 IEEE International Geoscience and Remote Sensing Symposium*, pages 7066–7070. IEEE, 2024.
- [56] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PmLR, 2021.
- [57] Colorado J Reed, Ritwik Gupta, Shufan Li, Sarah Brockman, Christopher Funk, Brian Clipp, Kurt Keutzer, Salvatore Candido, Matt Uyttendaele, and Trevor Darrell. Scale-mae: A scale-aware masked autoencoder for multiscale geospatial representation learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4088–4099, 2023.
- [58] Esther Rolf, Konstantin Klemmer, Caleb Robinson, and Hannah Kerner. Position: Mission critical – satellite data is a distinct modality in machine learning. *ICML*, 2024.
- [59] Michael Schmitt, Lloyd Haydn Hughes, Chunping Qiu, and Xiao Xiang Zhu. Sen12ms—a curated dataset of georeferenced multi-spectral sentinel-1/2 imagery for deep learning and data fusion. *arXiv preprint arXiv:1906.07789*, 2019.
- [60] Christoph Schuhmann, Romain Beaumont, Richard Vencu, Cade Gordon, Ross Wightman, Mehdi Cherti, Theo Coombes, Aarush Katta, Clayton Mullis, Mitchell Wortsman, et al. Laion-5b: An open large-scale dataset for training next generation image-text models. *Advances in neural information processing systems*, 35:25278–25294, 2022.

- [61] Piyush Sharma, Nan Ding, Sebastian Goodman, and Radu Soricut. Conceptual captions: A cleaned, hypernymed, image alt-text dataset for automatic image captioning. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2556–2565. Association for Computational Linguistics, 2018.
- [62] Naomi Simumba, Nils Lehmann, Paolo Fraccaro, Hamed Alemohammad, Geeth De Mel, Salman Khan, Manil Maskey, Nicolas Longepe, Xiao Xiang Zhu, Hannah Kerner, et al. Geobench-2: From performance to capability, rethinking evaluation in geospatial ai. *arXiv preprint arXiv:2511.15658*, 2025.
- [63] Sagar Soni, Akshay Dudhane, Hiyam Debary, Mustansar Fiaz, Muhammad Akhtar Munir, Muhammad Sohail Danish, Paolo Fraccaro, Campbell D Watson, Levente J Klein, Fahad Shahbaz Khan, et al. Earthdial: Turning multi-sensory earth observations to interactive dialogues. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 14303–14313, 2025.
- [64] Aarohi Srivastava, Abhinav Rastogi, Abhishek Rao, et al. Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. *Transactions on Machine Learning Research*, 2023.
- [65] Adam J Stewart, Caleb Robinson, Isaac A Corley, Anthony Ortiz, Juan M Lavista Ferres, and Arindam Banerjee. Torchgeo: deep learning with geospatial data. *ACM Transactions on Spatial Algorithms and Systems*, 11(4):1–28, 2025.
- [66] Xian Sun, Peijin Wang, Zhiyuan Yan, Feng Xu, Ruiping Wang, Wenhui Diao, Jin Chen, Jihao Li, Yingchao Feng, Tao Xu, et al. FairIm: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery. *ISPRS Journal of Photogrammetry and Remote Sensing*, 184:116–130, 2022.
- [67] Chao Tao, Ji Qi, Guo Zhang, Qing Zhu, Weipeng Lu, and Haifeng Li. Tov: The original vision model for optical remote sensing image understanding via self-supervised learning. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 16:4916–4930, 2023.
- [68] Gabriel Tseng, Ruben Cartuyvels, Ivan Zvonkov, Mirali Purohit, David Rolnick, and Hannah Kerner. Galileo: Learning global & local features of many remote sensing modalities. In *Proceedings of the International Conference on Machine Learning*, 2025.
- [69] Leonard Waldmann, Ando Shah, Yi Wang, Nils Lehmann, Adam Stewart, Zhitong Xiong, Xiao Xiang Zhu, Stefan Bauer, and John Chuang. Panopticon: Advancing any-sensor foundation models for earth observation. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 2204–2214, 2025.
- [70] Fengxiang Wang, Hongzhen Wang, Di Wang, Zonghao Guo, Zhenyu Zhong, Long Lan, Wenjing Yang, and Jing Zhang. Harnessing massive satellite imagery with efficient masked image modeling. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 6935–6947, 2025.
- [71] Yi Wang, Nassim Ait Ali Braham, Zhitong Xiong, Chenying Liu, Conrad M Albrecht, and Xiao Xiang Zhu. Ssl4eo-s12: A large-scale multimodal, multitemporal dataset for self-supervised learning in earth observation [software and data sets]. *IEEE Geoscience and Remote Sensing Magazine*, 11(3):98–106, 2023.
- [72] Gui-Song Xia, Jingwen Hu, Fan Hu, Baoguang Shi, Xiang Bai, Yanfei Zhong, Liangpei Zhang, and Xiaoqiang Lu. Aid: A benchmark data set for performance evaluation of aerial scene classification. *IEEE Transactions on Geoscience and Remote Sensing*, 55(7):3965–3981, 2017.
- [73] Gui-Song Xia, Xiang Bai, Jian Ding, Zhen Zhu, Serge Belongie, Jiebo Luo, Mihai Datcu, Marcello Pelillo, and Liangpei Zhang. Dota: A large-scale dataset for object detection in aerial images. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3974–3983, 2018.

- [74] Aoran Xiao, Weihao Xuan, Junjue Wang, et al. Foundation models for remote sensing and earth observation: A survey. *IEEE Geoscience and Remote Sensing Magazine*, 2025.
- [75] Zhitong Xiong, Yi Wang, Fahong Zhang, Adam J Stewart, Joelle Hanna, Damian Borth, Ioannis Papoutsis, Bertrand Le Saux, Gustau Camps-Valls, and Xiao Xiang Zhu. Neural plasticity-inspired multimodal foundation model for earth observation. *arXiv preprint arXiv:2403.15356*, 2024.
- [76] Yi Yang and Shawn Newsam. Bag-of-visual-words and spatial extensions for land-use classification. In *ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (ACM GIS)*, pages 270–279, 2010.
- [77] Xiaohua Zhai, Joan Puigcerver, Alexander Kolesnikov, Pierre Ruysen, Carlos Riquelme, Mario Lucic, Josip Djolonga, Andre Susano Pinto, Maxim Neumann, Alexey Dosovitskiy, et al. A large-scale study of representation learning with the visual task adaptation benchmark. *arXiv preprint arXiv:1910.04867*, 2019.
- [78] Mingming Zhang, Qingjie Liu, and Yunhong Wang. Ctxmim: Context-enhanced masked image modeling for remote sensing image understanding. *ACM Transactions on Multimedia Computing, Communications and Applications*, 21(12):1–22, 2025.
- [79] Wei Zhang, Miaoxin Cai, Tong Zhang, Yin Zhuang, and Xuerui Mao. Earthgpt: A universal multimodal large language model for multisensor image comprehension in remote sensing domain. *IEEE Transactions on Geoscience and Remote Sensing*, 62:1–20, 2024.
- [80] Zilun Zhang, Tiancheng Zhao, Yulong Guo, and Jianwei Yin. Rs5m and georsclip: A large-scale vision-language dataset and a large vision-language model for remote sensing. *IEEE Transactions on Geoscience and Remote Sensing*, 62:1–23, 2024.

A Reproducibility

The supplementary repository contains the 152-paper corpus, extraction prompts, normalization code, harvested results, and figure code; a top-level `Makefile` regenerates everything end-to-end. We keep this appendix short and let the repository speak for the details.

B Extraction Pipeline

We extract structured records directly from the LaTeX sources of the arXiv papers using Claude Opus 4.7 and GPT 5.5 Codex, and we run the same pipeline on the remaining PDF-only papers after converting them to structured markdown with Docling [47]. The first pass writes the structured fields we analyze; a second LLM review pass flags disagreements for manual inspection. We normalize names to deduplicate repeated models, benchmarks, and pretraining datasets, then manually verify all extracted fields against the source papers. We also remove labels that are not held-out evaluation sets and generic dataset descriptions that do not name a specific pretraining corpus. The normalization code and validator ship with the released code.

C Weight & Code Release Audit

Each paper’s weight-release flag is set by extracting the explicit claim in the PDF or LaTeX source and, when that is unclear, checking whether the release is hosted publicly elsewhere. Pointers to the pretraining *dataset* or to a HuggingFace dataset entry do not count.

The corpus then splits as 93 release weights, 26 explicitly do not, and 33 remain uncertain. In total, $59/152 = 39\%$ release no model weights. A further $29/152 = 19\%$ ship a public code repository with no released model artifact.

Table 1: **Top-10 cases where the same evaluation gives different scores.** Each row groups reports for the same model checkpoint, benchmark, metric, and stated protocol. *Min* and *max* are the lowest and highest reported scores. # reports is the number of papers reporting that same nominal evaluation. Papers are ordered from min→max. The last column gives likely unreported setup choice behind the gap.

Model	Benchmark	Metric / eval	# reports	min	max	Papers	Likely missing setup detail
Scale-MAE [57]	NWPU-RESISC45 [13]	acc. / linear	3	33.0	89.6	MA3E [44]; Scale-MAE [57]; DOFA [75]	Linear-probe recipe for the released ViT-L checkpoint: optimizer, head LR, and 224 vs. 256 eval crop.
AnySat [2]	m-Cashew [42]	mIoU / FT	2	27.2	80.4	DOFA [75]; OlmoEarth [4]	GEO-Bench fine-tune pipeline: optimizer, schedule, decoder, and loaded AnySat checkpoint.
TOV [67]	NWPU-RESISC45 [13]	acc. / FT	3	46.7	92.6	CtxMIM [78]; SelectiveMAE [70]; GeRSP [34]	Fine-tune recipe on a third-party backbone; CtxMIM reruns baselines under one SGD recipe far below source-reported FT scores.
GPT-4o	UCMerced [76]	acc. / 0-shot	2	43.5	88.8	RingMo-Agent [32]; EarthDial [63]	Prompt template, class verbalizer, API snapshot, and temperature; neither paper fully discloses these choices.
SeCo [50]	NWPU-RESISC45 [13]	acc. / FT	5	61.5	92.9	CtxMIM [78]; SelectiveMAE [70]; GeRSP [34]; EarthGPT [79]; MA3E [44]	Same CtxMIM-recipe issue as TOV; the four non-CtxMIM reports cluster at 89.6–92.9.
AnySat [2]	m-BigEarthNet [42]	F1 / kNN	3	54.5	85.3	OlmoEarth [4]; TerraFM [16]; Galileo [68]	GEO-Bench kNN-probe pipeline; TerraFM and OlmoEarth cluster near 54 pts, while Galileo reports 30+ pts higher.
CROMA [26]	m-Cashew [42]	mIoU / linear	3	31.8	62.2	TerraFM [16]; Panopticon [69]; SatDiFuser [35]	GEO-Bench UperNet probing; within-paper offsets may be fair locally but block cross-paper comparison.
Satlas [3]	m-BigEarthNet [42]	F1 / kNN	3	51.9	82.1	TerraFM [16]; OlmoEarth [4]; Galileo [68]	Same kNN-probe issue as AnySat; Galileo reports a much higher score than the other two papers.
DOFA [75]	m-Cashew [42]	mIoU / linear	3	27.7	56.4	TerraFM [16]; SatDiFuser [35]; Panopticon [69]	GEO-Bench probing pipeline; same likely setup issue as the CROMA row.
DOFA [75]	m-SACropType [42]	mIoU / linear	3	25.4	51.3	TerraFM [16]; SatDiFuser [35]; Panopticon [69]	GEO-Bench probing pipeline; the same three reproducing papers disagree on a second task.

D Reported-number divergence: harvest details

Table 1 groups reported numbers by the strictest combination we can extract from each source paper: same model, same benchmark, same metric, same evaluation regime (linear / kNN / fine-tune / zero-shot / few-shot), and same training-fraction bucket. Anything outside that combination is unpinned, including the linear-probe recipe, the fine-tune recipe used to rerun baselines, the GEO-Bench probing pipeline used to evaluate a third-party backbone, and the prompt / API version / temperature for vision-language models. The source papers do not disclose those choices. The remaining filters (we keep only full-train results, drop classical-ML baselines, and exclude detection benchmarks where “mAP” conflates different definitions, e.g., mAP, mAP@50, and oriented-object-detection mAP) live in the released code.